LEARNING HUB MASTER SERIES ◆ VOLUME 06

Diagonalization & Matrix Powers

The Eigenvector Factorization $A = X\Lambda X^{-1}$, Discrete Iterations, Markov Steady States, Continuous Differential Flows e^{At}, and the Jordan Canonical Form

Theoretical Core: Spectral Diagonalization Theorem ◆ Multiplicity Criteria (AM vs. GM)
Discrete Dynamics: Matrix Powers A^k ◆ Fibonacci Companion Matrix ◆ Markov Chains
Continuous Dynamics: Matrix Exponential e^{At} ◆ Coupled ODE Decoupling ◆ Phase Portraits
Canonical Forms: Similarity Invariants ◆ Jordan Blocks ◆ Nilpotent Shear Operators

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Author Reference: Synthesized from Gilbert Strang, *Introduction to Linear Algebra* (5th Edition, Chapters 6 & 7) and Kenji Hiranabe, *The Art of Linear Algebra*.
Target Scope: Engineering Mathematics, Physics, Advanced Computer Science & Applied Dynamical Systems.
Standard Edition: English Translation & Analytical Reference Handbook (100% Rigorous Formulations).

Table of Contents & Structural Syllabus

Part 1: The Diagonalization Theorem & Spectral Factorization

1.1 The Fundamental Identity $AX = X\Lambda$ and Factoring $A = X\Lambda X^{-1}$ — Theory
1.2 Invertibility of the Eigenvector Matrix X & Linear Independence — Proof
1.3 Distinct Eigenvalues Guarantee Universal Diagonalizability — Theorem
1.4 Change of Basis: Standard Coordinates to Uncoupled Eigenbasis — Figure 6.1

Part 2: Distinct vs. Repeated Eigenvalues & Defective Matrices

2.1 Algebraic Multiplicity (AM) vs. Geometric Multiplicity (GM) — Definition
2.2 The Fundamental Multiplicity Inequality: $1 \le GM(\lambda) \le AM(\lambda)$ — Criterion
2.3 Defective Matrices & Geometric Shear Actions — Figure 6.2
2.4 Non-Defective Matrices with Repeated Eigenvalues (Full Eigenspaces) — Analysis

Part 3: Powers of a Matrix & Discrete Difference Systems

3.1 High Matrix Powers via Factorization: $A^k = X\Lambda^k X^{-1}$ — Calculation
3.2 Spectral Radius $\rho(A)$ and Asymptotic Stability ($A^k \to O$) — Theorem
3.3 The Fibonacci Recurrence as a First-Order Matrix Difference System — Derivation
3.4 Markov Transition Matrices, Column Stochasticity & Perron-Frobenius — Figure 6.3

Part 4: Systems of Linear Differential Equations & The Matrix Exponential

4.1 Coupled First-Order ODE Systems: $du/dt = Au$ — Decoupling
4.2 The Matrix Exponential e^{At}: Power Series & Spectral Construction — Identity
4.3 Asymptotic Stability Criteria: $Re(\lambda_i) < 0$ and the Trace-Determinant Plane — Phase Plane

Part 5: Similarity Transformations, Invariants & Jordan Canonical Form

5.1 Similarity Transformations $B = P^{-1}AP$ as Basis Shifts — Geometry
5.2 Complete Inventory of Similarity Invariants (Trace, Det, Charpoly, Rank) — Properties
5.3 Jordan Canonical Form $J = M^{-1}AM$ for Defective Matrices — Theorem
5.4 Nilpotent Shift Operators N and Generalized Eigenvector Chains — Figure 6.4

Part 6: Fast-Track Calculation Shortcut Compendium

6.1 Instant 2×2 Eigenvalue Quick-Formula & Discriminant Test — Fast-Track 6.1
6.2 Rapid 2×2 Eigenvector Construction via Row Adjugate — Fast-Track 6.2
6.3 Instant Steady-State Markov Distribution without Inverting X — Fast-Track 6.3
6.4 Matrix Powers of Shifted / Rank-1 Perturbations — Fast-Track 6.4
6.5 Nilpotent Upper-Triangular Toeplitz Exponential — Fast-Track 6.5
6.6 Cayley-Hamilton Degree Reduction for High Matrix Powers — Fast-Track 6.6

Part 7: Step-by-Step Worked Examination Problems

Problem 6.1: Full Diagonalization & Exact High-Power Evaluation (A^{10}) — Worked
Problem 6.2: Coupled Difference Equation: Fibonacci Sequence & Binet's Formula — Worked
Problem 6.3: Markov System: Urban-Suburban Population Migration Dynamics — Worked
Problem 6.4: System of Linear Differential Equations via Matrix Exponential — Worked
Problem 6.5: Defective Matrix Analysis & Jordan Canonical Form Construction — Worked

Part 1: The Diagonalization Theorem & Spectral Factorization

1.1 The Fundamental Matrix Equation AX = XΛ

The central factorization of matrix linear algebra expresses a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ in terms of its eigenvalues and eigenvectors. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of **A**, and let $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ be the corresponding eigenvectors satisfying the defining eigenvalue equation $\mathbf{A}\mathbf{x}_i = \lambda_i \mathbf{x}_i$.

When we assemble the *n* eigenvectors as columns of a square matrix $\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \ldots \quad \mathbf{x}_n]$, and multiply **A** by **X**, the action of **A** on each column proceeds independently:

$$\mathbf{AX} = \mathbf{A}[\mathbf{x}_1 \quad \mathbf{x}_2 \quad \ldots \quad \mathbf{x}_n] = [\mathbf{A}\mathbf{x}_1 \quad \mathbf{A}\mathbf{x}_2 \quad \ldots \quad \mathbf{A}\mathbf{x}_n] = [\lambda_1\mathbf{x}_1 \quad \lambda_2\mathbf{x}_2 \quad \ldots \quad \lambda_n\mathbf{x}_n]$$

Crucially, post-multiplying **X** by a diagonal matrix $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ precisely scales each column *i* of **X** by λ_i:

$$\mathbf{X\Lambda} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \ldots \quad \mathbf{x}_n] \begin{vmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{vmatrix} = [\lambda_1\mathbf{x}_1 \quad \lambda_2\mathbf{x}_2 \quad \ldots \quad \lambda_n\mathbf{x}_n]$$

◆ Theorem 1.1: The Diagonalization Theorem

An $n \times n$ matrix **A** is diagonalizable if and only if it possesses **n linearly independent eigenvectors**. If such eigenvectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$ exist, the eigenvector matrix **X** is invertible ($\det(\mathbf{X}) \neq 0$), and **A** can be factored into the similarity product:

$$\mathbf{A} = \mathbf{X}\,\mathbf{\Lambda}\,\mathbf{X}^{-1} \quad \Leftrightarrow \quad \mathbf{\Lambda} = \mathbf{X}^{-1}\,\mathbf{A}\,\mathbf{X}$$

1.2 Invertibility of X & Independence of Eigenvectors

The factorization $\mathbf{A} = \mathbf{X\Lambda X}^{-1}$ is impossible unless $\mathbf{X}^{-1}$ exists. By the Invertible Matrix Theorem, **X** is invertible if and only if its columns $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are linearly independent in $\mathbb{R}^n$ (or $\mathbb{C}^n$). If **A** has fewer than *n* independent eigenvectors, **X** is singular ($\mathrm{rank}(\mathbf{X}) < n$), and **A** cannot be diagonalized.

◆ Formal Proof: Distinct Eigenvalues Guarantee Linear Independence

Statement: If $\lambda_1, \lambda_2, \ldots, \lambda_k$ are distinct eigenvalues of $\mathbf{A}$ with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$, then $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is a linearly independent set.

Proof by Induction on k:

Base Step (k = 1): Since $\mathbf{x}_1 \neq \mathbf{0}$ by definition of an eigenvector, $c_1\mathbf{x}_1 = \mathbf{0} \Rightarrow c_1 = 0$. True.

Inductive Hypothesis: Assume any set of *k - 1* eigenvectors with distinct eigenvalues is linearly independent.

Inductive Step: Suppose there exist scalars $c_1, \ldots, c_k$ such that:

$$\text{(Eq. 1)} \quad c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \ldots + c_{k-1}\mathbf{x}_{k-1} + c_k\mathbf{x}_k = \mathbf{0}$$

Multiply Eq. 1 by the matrix $\mathbf{A}$ on the left:

$$\text{(Eq. 2)} \quad c_1\lambda_1\mathbf{x}_1 + c_2\lambda_2\mathbf{x}_2 + \ldots + c_{k-1}\lambda_{k-1}\mathbf{x}_{k-1} + c_k\lambda_k\mathbf{x}_k = \mathbf{0}$$

Next, multiply Eq. 1 by the scalar λ_k:

$$\text{(Eq. 3)} \quad c_1\lambda_k\mathbf{x}_1 + c_2\lambda_k\mathbf{x}_2 + \ldots + c_{k-1}\lambda_k\mathbf{x}_{k-1} + c_k\lambda_k\mathbf{x}_k = \mathbf{0}$$

Subtract Eq. 3 from Eq. 2 to eliminate the final term $c_k\lambda_k\mathbf{x}_k$:

$$c_1(\lambda_1 - \lambda_k)\mathbf{x}_1 + c_2(\lambda_2 - \lambda_k)\mathbf{x}_2 + \ldots + c_{k-1}(\lambda_{k-1} - \lambda_k)\mathbf{x}_{k-1} = \mathbf{0}$$

By the induction hypothesis, $\{\mathbf{x}_1, \ldots, \mathbf{x}_{k-1}\}$ is linearly independent. Hence each coefficient must vanish:

$$c_i(\lambda_i - \lambda_k) = 0 \quad \text{for all } i = 1, 2, \ldots, k - 1.$$

Because all eigenvalues are distinct by assumption, $\lambda_i - \lambda_k \neq 0$ for all $i \neq k$. Therefore, $c_1 = c_2 = \ldots = c_{k-1} = 0$. Substituting these into Eq. 1 leaves $c_k\mathbf{x}_k = \mathbf{0}$. Since $\mathbf{x}_k \neq \mathbf{0}$, $c_k = 0$ as well. All scalars c_i are zero, establishing linear independence for all *k*. ■

◆ Geometric Meaning: The Diagonalization Pipeline

Geometrically, the similarity transformation $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$ is a three-stage coordinate pipeline:

1. **$\mathbf{X}^{-1}$ (Analysis):** Decomposes an arbitrary vector $\mathbf{u}$ from the standard coordinate system into its eigenbasis coordinates: $\mathbf{c} = \mathbf{X}^{-1}\mathbf{u}$.
2. **$\mathbf{\Lambda}$ (Decoupled Scaling):** Scales each eigenbasis component c_i independently by its corresponding eigenvalue λ_i. There is zero cross-coupling between different modes.
3. **$\mathbf{X}$ (Synthesis):** Reassembles the scaled eigenbasis components back into standard spatial coordinates: $\mathbf{Au} = \mathbf{X}(\mathbf{\Lambda c}) = \sum c_i \lambda_i \mathbf{x}_i$.

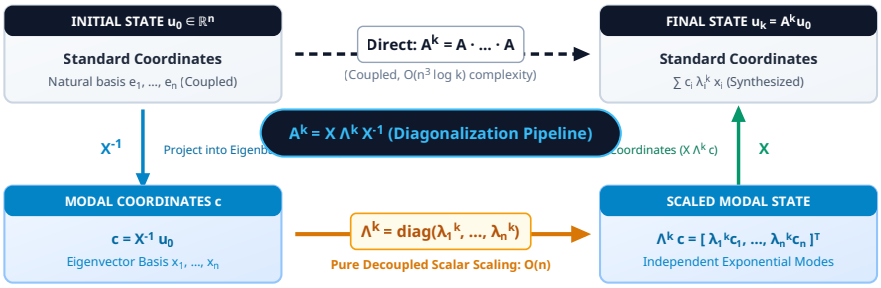


Figure 6.1: *The Change-of-Basis Diagonalization Pipeline. Direct matrix multiplication $\mathbf{A}^k$ in the coupled standard basis is computationally expensive. Factoring via $\mathbf{X}^{-1}$ projects coordinates into the eigenbasis where the action $\mathbf{\Lambda}^k$ reduces to trivial decoupled scalar exponentiation (λ_i^k), before $\mathbf{X}$ reconstructs the output in standard space.*

Part 2: Distinct vs. Repeated Eigenvalues & Defective Matrices

2.1 Algebraic Multiplicity (AM) vs. Geometric Multiplicity (GM)

When the characteristic equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ has repeated roots, diagonalizability is no longer guaranteed. To resolve whether a matrix with repeated eigenvalues can be diagonalized, we strictly distinguish between two fundamental forms of multiplicity:

- **Algebraic Multiplicity, AM(λ):** The multiplicity of λ as a root of the characteristic polynomial $p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = 0$. If $p(\lambda) = (\lambda - \lambda_0)^m q(\lambda)$ with $q(\lambda_0) \neq 0$, then $AM(\lambda_0) = m$.
- **Geometric Multiplicity, GM(λ):** The maximum number of linearly independent eigenvectors associated with λ. Equivalently, it is the dimension of the eigenspace E_λ:

$$GM(\lambda) = \dim(\text{null}(\mathbf{A} - \lambda\mathbf{I})) = n - \text{rank}(\mathbf{A} - \lambda\mathbf{I})$$

◆ Theorem 2.1: The Fundamental Multiplicity Inequality

For every eigenvalue λ of any square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$:

$$\mathbf{1 \leq GM(\lambda) \leq AM(\lambda)}$$

The Diagonalizability Test: An $n \times n$ matrix **A** is diagonalizable if and only if:

$$\mathbf{GM(\lambda_i) = AM(\lambda_i)} \quad \textbf{for every eigenvalue } \boldsymbol{\lambda_i}$$

Equivalently, the sum of the geometric multiplicities over all distinct eigenvalues must equal n: $\sum GM(\lambda_i) = n$.

2.2 Defective Matrices & Coordinate Shear

If for any eigenvalue λ, we find $GM(\lambda) < AM(\lambda)$, the matrix is termed **defective**. A defective matrix lacks a full set of n independent eigenvectors; its eigenspace is short of dimensions, making the construction of an invertible eigenvector matrix **X** impossible.

⚠ Warning: The Two Classic Archetypes of Repeated Eigenvalues

Consider two 2×2 matrices with the identical characteristic polynomial $p(\lambda) = (\lambda - 3)^2 \Rightarrow \lambda = 3$ with AM = 2:

Archetype 1 (Non-Defective / Diagonal): $\mathbf{A} = \begin{vmatrix} 3 & 0 \\ 0 & 3 \end{vmatrix}$.

Eigenspace: $\mathbf{A} - 3\mathbf{I} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$. Rank = 0 ⇒ GM = 2 - 0 = 2. Here GM = AM = 2. **A** has two independent eigenvectors $[1, 0]^T$ and $[0, 1]^T$. It is already fully diagonal!

Archetype 2 (Defective / Jordan Block): $\mathbf{B} = \begin{vmatrix} 3 & 1 \\ 0 & 3 \end{vmatrix}$.

Eigenspace: $\mathbf{B} - 3\mathbf{I} = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}$. Rank = 1 ⇒ GM = 2 - 1 = 1. Here GM = 1 < AM = 2.

The eigenspace is only 1-dimensional: span($[1, 0]^T$). The matrix **B** is defective and cannot be diagonalized under any similarity transformation!

Geometrically, a diagonalizable matrix scales space along independent eigenvector axes. A defective matrix introduces a **shear deformation**: parallel coordinate lines slide past one another, preventing the formation of an invariant orthogonal or independent coordinate frame.

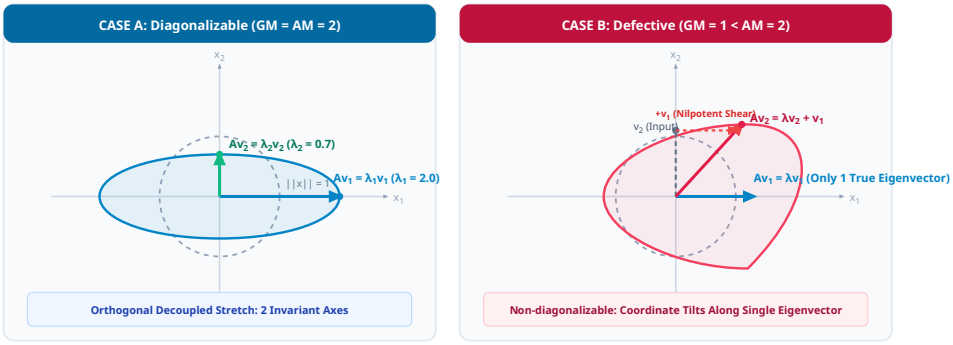


Figure 6.2: *Coordinate Transformation: Diagonal Scaling vs. Jordan Defective Shear. Case A (left) possesses 2 independent eigenvectors (GM = AM = 2), producing a clean elliptical stretch along invariant axes. Case B (right) has only 1 independent eigenvector (GM = 1 < AM = 2), causing a shear distortion where coordinates slide horizontally along the single invariant vector direction.*

2.3 Non-Defective Matrices with Repeated Eigenvalues (Full Eigenspaces)

A repeated eigenvalue does **not** automatically imply defectiveness. A matrix with repeated eigenvalues is fully diagonalizable if and only if the eigenspace for each repeated eigenvalue has dimension matching its algebraic multiplicity: dim(null(**A** - λ**I**)) = AM(λ).

Canonical Families of Non-Defective Matrices with Repeated Roots:

- **Scalar Identity Multiples:** **A** = c**I**. The characteristic polynomial is $(\lambda - c)^n = 0$ (AM = n). Here **A** - c**I** = **O**, so null(**A** - c**I**) = $\mathbb{R}^n$ (GM = n). Every non-zero vector in $\mathbb{R}^n$ is an eigenvector, and the eigenspace has full dimension *n*.
- **Projection Matrices:** Any projection matrix **P** (satisfying $\mathbf{P}^2 = \mathbf{P}$) has eigenvalues λ ∈ {0, 1}. If rank(**P**) = r, then λ = 1 has AM = GM = r (the column space / projection subspace), and λ = 0 has AM = GM = n - r (the nullspace). Projection matrices are universally non-defective and diagonalizable.
- **Real Symmetric Matrices:** As established in Volume 07, every real symmetric matrix $\mathbf{S} = \mathbf{S}^T$ has GM(λ) = AM(λ) for all eigenvalues, guaranteeing universal orthogonal diagonalizability without exception.

Part 3: Powers of a Matrix & Discrete Difference Systems

3.1 High Matrix Powers via Factorization

Direct multiplication to compute $\mathbf{A}^k$ requires $k - 1$ full matrix products, demanding $O(k\, n^3)$ operations. If $\mathbf{A}$ is diagonalizable, the factorization $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$ transforms this into a trivial calculation:

$$\mathbf{A}^k = (\mathbf{X}\,\mathbf{\Lambda}\,\mathbf{X}^{-1})(\mathbf{X}\,\mathbf{\Lambda}\,\mathbf{X}^{-1}) \dots (\mathbf{X}\,\mathbf{\Lambda}\,\mathbf{X}^{-1}) = \mathbf{X}\,\mathbf{\Lambda}\,(\mathbf{X}^{-1}\,\mathbf{X})\,\mathbf{\Lambda} \dots (\mathbf{X}^{-1}\,\mathbf{X})\,\mathbf{\Lambda}\,\mathbf{X}^{-1} = \mathbf{X}\,\mathbf{\Lambda}^k\,\mathbf{X}^{-1}$$

Because $\mathbf{\Lambda}$ is diagonal, raising it to the power k merely raises its diagonal entries to the power k: $\mathbf{\Lambda}^k = \mathrm{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k)$. The entire power calculation reduces to:

$$\mathbf{A}^k = \mathbf{X} \begin{vmatrix} \lambda_1^k & 0 & \dots & 0 \\ 0 & \lambda_2^k & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n^k \end{vmatrix} \mathbf{X}^{-1}$$

3.2 Discrete Difference Equations: $\mathbf{u}_{k+1} = \mathbf{A}\mathbf{u}_k$

In discrete dynamical systems, an initial state vector $\mathbf{u}_0$ evolves according to the linear recurrence $\mathbf{u}_{k+1} = \mathbf{A}\mathbf{u}_k$. Successive substitution yields $\mathbf{u}_k = \mathbf{A}^k\,\mathbf{u}_0$. Expanding $\mathbf{u}_0$ in the basis of eigenvectors:

$$\mathbf{u}_0 = c_1\,\mathbf{x}_1 + c_2\,\mathbf{x}_2 + \dots + c_n\,\mathbf{x}_n = \mathbf{X}\,\mathbf{c} \quad \Rightarrow \quad \mathbf{c} = \mathbf{X}^{-1}\,\mathbf{u}_0$$

Multiplying by $\mathbf{A}^k$ yields the fundamental modal decomposition of the trajectory:

$$\mathbf{u}_k = \mathbf{A}^k\,\mathbf{u}_0 = c_1\,\lambda_1^k\,\mathbf{x}_1 + c_2\,\lambda_2^k\,\mathbf{x}_2 + \dots + c_n\,\lambda_n^k\,\mathbf{x}_n$$

◆ Theorem 3.1: Asymptotic Stability & Spectral Radius

The asymptotic long-term behavior of $\mathbf{u}_k = \mathbf{A}^k\,\mathbf{u}_0$ as $k \to \infty$ is governed strictly by the magnitudes of its eigenvalues $|\lambda_i|$:

- **Asymptotically Stable (Decay to 0):** $\mathbf{A}^k \to \mathbf{O}$ as $k \to \infty$ for all $\mathbf{u}_0$ if and only if the **spectral radius** $\rho(\mathbf{A}) = \max_i |\lambda_i| < 1$.
- **Neutral / Steady State:** If the dominant eigenvalue is $\lambda_1 = 1$ and all other $|\lambda_i| < 1$, then as $k \to \infty$, $\mathbf{u}_k \to c_1\,\mathbf{x}_1$. The system converges to a steady state.
- **Unstable (Blow Up):** If any $|\lambda_i| > 1$, trajectories grow exponentially without bound along span($\mathbf{x}_i$).

3.3 Markov Transition Matrices & Steady-State Convergence

A square matrix $\mathbf{M}$ is a **Markov transition matrix** (or column-stochastic matrix) if:

1. All entries are non-negative: $m_{ij} \geq 0$ for all i, j.
2. Every column sums to exactly 1: $\sum_{i=1}^{n} m_{ij} = 1$ for all j.

◆ Theorem & Proof: The Universal Steady State $\lambda_1 = 1$

Statement: Every Markov matrix $\mathbf{M}$ possesses an eigenvalue $\lambda = 1$.
Proof: Let $\mathbf{1} = [1, 1, \ldots, 1]^T$ be the vector of all ones. Because each column of $\mathbf{M}$ sums to 1, the row vector $\mathbf{1}^T \mathbf{M} = \mathbf{1}^T$, which means $\mathbf{M}^T \mathbf{1} = \mathbf{1} = 1 \cdot \mathbf{1}$.
This proves that $\lambda = 1$ is an eigenvalue of the transpose $\mathbf{M}^T$ with eigenvector $\mathbf{1}$.
Since any matrix and its transpose share the identical characteristic polynomial $\det(\mathbf{M}^T - \lambda \mathbf{I}) = \det(\mathbf{M} - \lambda \mathbf{I}) = 0$, $\mathbf{M}$ must also possess the eigenvalue $\lambda_1 = 1$. ■

By the **Perron-Frobenius Theorem**, if all entries of $\mathbf{M}$ (or a power $\mathbf{M}^p$) are strictly positive, $\lambda_1 = 1$ is a strictly dominant, simple eigenvalue: all other eigenvalues satisfy $|\lambda_i| < 1$. Consequently, as $k \to \infty$:

$$\mathbf{u}_k = \mathbf{M}^k \mathbf{u}_0 = c_1 (1)^k \mathbf{x}_1 + \sum_{i=2}^{n} c_i \lambda_i^k \mathbf{x}_i \quad \longrightarrow \quad c_1 \mathbf{x}_1 = \mathbf{u}_\infty$$

The system forgets its initial state $\mathbf{u}_0$ and asymptotically approaches the unique steady-state distribution $\mathbf{u}_\infty$.

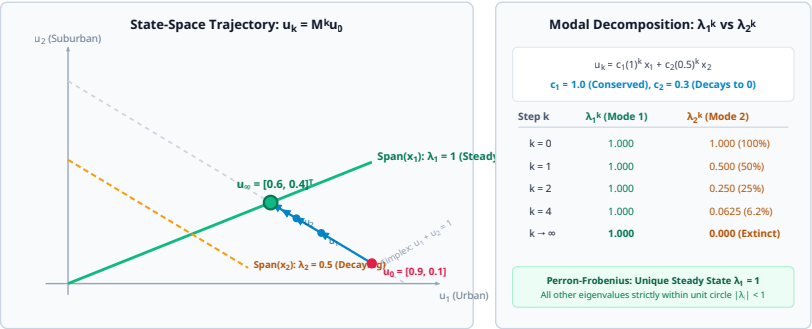


Figure 6.3: *Discrete Dynamical System Power Trajectory & Markov Steady-State. Starting from an arbitrary initial population distribution $\mathbf{u}_0 = [0.9, 0.1]^T$, the decaying mode $\lambda_2^k = (0.5)^k$ contracts exponentially at each iteration, pulling the state directly onto the dominant eigenspace span($\mathbf{x}_1$) and converging to the unique steady state $\mathbf{u}_\infty = [0.6, 0.4]^T$.*

Part 4: Systems of Differential Equations & The Matrix Exponential

4.1 Decoupling Systems of Linear Differential Equations

Consider a continuous dynamical system governed by a system of n coupled first-order linear ordinary differential equations:

$$\frac{d\mathbf{u}}{dt} = \mathbf{A}\,\mathbf{u}(t), \quad \text{with initial condition } \mathbf{u}(0) = \mathbf{u}_0$$

Because $\mathbf{A}$ is non-diagonal, the derivatives du_i/dt mix components of $\mathbf{u}$. If $\mathbf{A}$ is diagonalizable ($\mathbf{A} = \mathbf{X}\Lambda\mathbf{X}^{-1}$), we decouple the system by introducing modal coordinates $\mathbf{v}(t)$ via the substitution $\mathbf{u}(t) = \mathbf{X}\,\mathbf{v}(t)$:

$$\mathbf{X}\frac{d\mathbf{v}}{dt} = \mathbf{A}\,\mathbf{X}\,\mathbf{v} \quad \Rightarrow \quad \frac{d\mathbf{v}}{dt} = \mathbf{X}^{-1}\mathbf{A}\,\mathbf{X}\,\mathbf{v} = \Lambda\,\mathbf{v}$$

Because Λ is diagonal, the system decouples into n independent scalar ODEs: $dv_i/dt = \lambda_i\, v_i$, with trivial solutions $v_i(t) = c_i\, e^{\lambda_i t}$. Transforming back to original coordinates yields:

$$\mathbf{u}(t) = \mathbf{X}\,\mathbf{v}(t) = c_1\, e^{\lambda_1 t}\,\mathbf{x}_1 + c_2\, e^{\lambda_2 t}\,\mathbf{x}_2 + \ldots + c_n\, e^{\lambda_n t}\,\mathbf{x}_n$$

4.2 The Matrix Exponential e^{At}

In exact analogy to the scalar ODE $du/dt = au \Rightarrow u(t) = e^{at}u_0$, the solution to the vector system is $\mathbf{u}(t) = e^{\mathbf{A}t}\,\mathbf{u}_0$, where the **matrix exponential** is defined by the globally convergent Taylor series:

$$e^{\mathbf{A}t} = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2!} + \frac{\mathbf{A}^3 t^3}{3!} + \ldots$$

◆ Theorem 4.1: Spectral Formula for the Matrix Exponential

If $\mathbf{A}$ is diagonalizable ($\mathbf{A} = \mathbf{X}\,\Lambda\,\mathbf{X}^{-1}$), substituting $\mathbf{A}^k = \mathbf{X}\,\Lambda^k\,\mathbf{X}^{-1}$ into the Taylor series yields:

$$e^{\mathbf{A}t} = \mathbf{X}\,e^{\Lambda t}\,\mathbf{X}^{-1} = \mathbf{X}\begin{vmatrix} e^{\lambda_1 t} & 0 & \ldots & 0 \\ 0 & e^{\lambda_2 t} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & e^{\lambda_n t} \end{vmatrix}\mathbf{X}^{-1}$$

Fundamental Properties of $e^{\mathbf{A}t}$:

1. $\frac{d}{dt}(e^{\mathbf{A}t}) = \mathbf{A}\,e^{\mathbf{A}t} = e^{\mathbf{A}t}\,\mathbf{A}$.
2. $e^{\mathbf{A}(t+s)} = e^{\mathbf{A}t}\,e^{\mathbf{A}s}$ and $(e^{\mathbf{A}t})^{-1} = e^{-\mathbf{A}t}$.
3. **Jacobi's Formula:** $\det(e^{\mathbf{A}t}) = e^{\text{trace}(\mathbf{A})t}$. Because $e^{\text{trace}(\mathbf{A})t} \neq 0$, $e^{\mathbf{A}t}$ is *always invertible* for any matrix $\mathbf{A}$ and any t!

4.3 Continuous Stability & The Trace-Determinant Plane

The solution $\mathbf{u}(t) = \sum c_i e^{\lambda_i t} \mathbf{x}_i$ decays to zero as $t \to \infty$ if and only if every exponential factor satisfies $e^{\lambda_i t} \to 0$. For complex eigenvalues $\lambda_i = \alpha_i + i\beta_i$, $|e^{\lambda_i t}| = e^{\alpha_i t}$.

◆ Theorem 4.2: Continuous Dynamical Stability Criteria

The linear system $du/dt = \mathbf{A}u$ is **asymptotically stable** ($\mathbf{u}(t) \to \mathbf{0}$ as $t \to \infty$) if and only if all eigenvalues have strictly negative real parts:

$$\text{Re}(\lambda_i) < 0 \quad \text{for all } i = 1, 2, \ldots, n$$

For a 2×2 real system with characteristic equation $\lambda^2 - \text{trace}(\mathbf{A})\lambda + \det(\mathbf{A}) = 0$, stability holds if and only if:

$$\text{trace}(A) < 0 \quad \text{and} \quad \det(A) > 0$$

Condition on Trace and Det	Discriminant $\Delta = \text{tr}^2 - 4\det$	Eigenvalues λ_1, λ_2	Phase Portrait Classification
$\det(\mathbf{A}) < 0$	$\Delta > 0$	Real, opposite signs ($\lambda_1 > 0$, $\lambda_2 < 0$)	**Saddle Point** (Always Unstable)
$\text{tr}(\mathbf{A}) < 0$, $\det(\mathbf{A}) > 0$	$\Delta \geq 0$	Real, both negative ($\lambda_2 \leq \lambda_1 < 0$)	**Stable Node** (Asymptotically Stable Sink)
$\text{tr}(\mathbf{A}) < 0$, $\det(\mathbf{A}) > 0$	$\Delta < 0$	Complex conjugate, $\text{Re}(\lambda) < 0$	**Stable Spiral Sink** (Damped Oscillation)
$\text{tr}(\mathbf{A}) = 0$, $\det(\mathbf{A}) > 0$	$\Delta < 0$	Pure imaginary $\lambda = \pm i\omega$	**Center** (Neutral Undamped Periodic Orbit)
$\text{tr}(\mathbf{A}) > 0$, $\det(\mathbf{A}) > 0$	$\Delta \geq 0$ or $\Delta < 0$	Real positive or $\text{Re}(\lambda) > 0$	**Unstable Node / Spiral Source**

Part 5: Similarity Transformations & Jordan Canonical Form

5.1 Similarity Transformations as Basis Changes

Two square matrices **A** and **B** are **similar** (written **B** ~ **A**) if there exists an invertible matrix **P** such that:

$$\mathbf{B} = \mathbf{P}^{-1}\,\mathbf{A}\,\mathbf{P}$$

A similarity transformation represents a change of basis for a linear operator T: V → V. If **A** is the matrix of T relative to standard basis B_1, and **P** is the transition matrix to a new basis B_2, then **B** is the matrix representing the exact same operator T relative to B_2.

◆ **Theorem 5.1: The Complete Inventory of Similarity Invariants**

If $\mathbf{B} = \mathbf{P}^{-1}\,\mathbf{A}\,\mathbf{P}$, then **A** and **B** share the identical intrinsic structural properties:

1. **Characteristic Polynomial:** $\det(\mathbf{B} - \lambda\mathbf{I}) = \det(\mathbf{P}^{-1}(\mathbf{A} - \lambda\mathbf{I})\mathbf{P}) = \det(\mathbf{P}^{-1})\det(\mathbf{A} - \lambda\mathbf{I})\det(\mathbf{P}) = \det(\mathbf{A} - \lambda\mathbf{I})$.
2. **Eigenvalues:** **A** and **B** have the exact same eigenvalues with identical algebraic multiplicities AM.
3. **Trace:** $\text{trace}(\mathbf{B}) = \text{trace}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \text{trace}(\mathbf{A}\mathbf{P}\mathbf{P}^{-1}) = \text{trace}(\mathbf{A})$.
4. **Determinant:** $\det(\mathbf{B}) = \det(\mathbf{P}^{-1})\det(\mathbf{A})\det(\mathbf{P}) = \det(\mathbf{A})$.
5. **Rank & Nullity:** $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$ and $\text{nullity}(\mathbf{B}) = \text{nullity}(\mathbf{A})$.
6. **Geometric Multiplicities:** $\dim(\text{null}(\mathbf{B} - \lambda\mathbf{I})) = \dim(\text{null}(\mathbf{A} - \lambda\mathbf{I}))$ for every λ.
7. **Minimal Polynomial:** $m_B(\lambda) = m_A(\lambda)$.

What Changes? The eigenvectors change: if $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, then $\mathbf{B}(\mathbf{P}^{-1}\mathbf{x}) = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}(\mathbf{P}^{-1}\mathbf{x}) = \mathbf{P}^{-1}\mathbf{A}\mathbf{x} = \lambda(\mathbf{P}^{-1}\mathbf{x})$. The eigenvectors of **B** are $\mathbf{y} = \mathbf{P}^{-1}\,\mathbf{x}$.

5.2 The Jordan Canonical Form for Defective Matrices

When a matrix is defective, it cannot be diagonalized into **Λ**. However, Jordan's Theorem guarantees that every square matrix is similar to an almost-diagonal matrix composed of block diagonals called **Jordan blocks**.

◆ Theorem 5.2: The Jordan Canonical Form

Every $n \times n$ complex matrix $\mathbf{A}$ is similar to a Jordan form $\mathbf{J} = \mathbf{M}^{-1}\mathbf{A}\mathbf{M}$:

$$\mathbf{J} = \operatorname{diag}(\mathbf{J}_1, \mathbf{J}_2, \ldots, \mathbf{J}_p) = \begin{vmatrix} \mathbf{J}_1 & \mathbf{O} & \cdots & \mathbf{O} \\ \mathbf{O} & \mathbf{J}_2 & \cdots & \mathbf{O} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{O} & \mathbf{O} & \cdots & \mathbf{J}_p \end{vmatrix}$$

Each Jordan block $\mathbf{J}_i$ has an eigenvalue λ on the diagonal and 1s on the superdiagonal:

$$\mathbf{J}_i(\lambda) = \lambda\,\mathbf{I}_m + \mathbf{N}_m = \begin{vmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{vmatrix}$$

Fundamental Rules of the Jordan Structure:

1. The total number of Jordan blocks corresponding to λ equals **GM(λ)**.
2. The sum of the sizes of all Jordan blocks corresponding to λ equals **AM(λ)**.
3. The size of the largest Jordan block for λ is the multiplicity of λ in the **minimal polynomial**.

5.3 Nilpotent Shift Operators & Jordan Chains

The matrix $\mathbf{N}$ with 1s on the superdiagonal is **nilpotent**: raising it to powers shifts the ones upward until $\mathbf{N}^m = \mathbf{O}$. The columns of the similarity matrix $\mathbf{M} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_m]$ form a **Jordan chain of generalized eigenvectors** satisfying:

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v}_1 = \mathbf{0}, \quad (\mathbf{A} - \lambda\mathbf{I})\mathbf{v}_2 = \mathbf{v}_1, \quad \ldots, \quad (\mathbf{A} - \lambda\mathbf{I})\mathbf{v}_m = \mathbf{v}_{m-1}$$

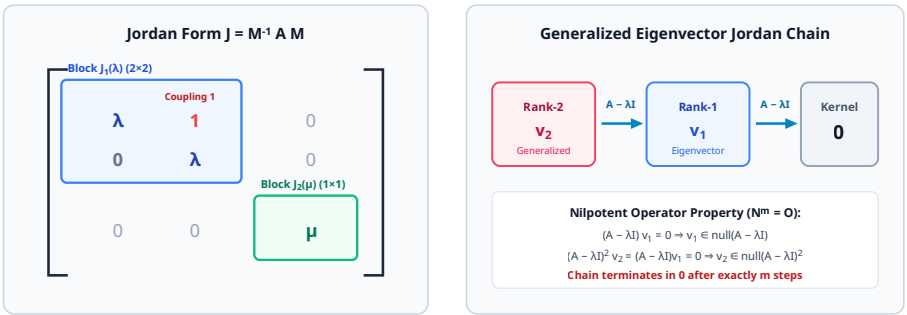


Figure 6.4: *Jordan Canonical Form and Generalized Eigenvector Chain. For a defective matrix, true eigenvectors span the kernel of ($\mathbf{A} - \lambda\mathbf{I}$). Generalized eigenvectors form a chain linking higher-order nullspaces null($(\mathbf{A} - \lambda\mathbf{I})^k$) down to the true eigenvector $\mathbf{v}_1$, which maps to zero under the nilpotent shift operator.*

Part 6: Fast-Track Calculation Shortcut Compendium

⚡ Fast-Track 6.1: Instant 2×2 Eigenvalue Quick-Formula & Discriminant

For any 2 × 2 matrix $\mathbf{A} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, never write out $\det(\mathbf{A} - \lambda\mathbf{I})$ from scratch. Immediately compute the trace and determinant:

$$\text{trace}(A) = a + d, \quad \det(A) = ad - bc \quad \Rightarrow \quad \lambda^2 - \text{trace}(A)\lambda + \det(A) = 0$$

The eigenvalues are given directly by:

$$\lambda_{1,2} = \frac{\text{trace}(A) \pm \sqrt{(\text{trace}(A))^2 - 4\det(A)}}{2}$$

Discriminant Shortcut: $\Delta = (\text{trace})^2 - 4\det = (a - d)^2 + 4bc$. If $\Delta > 0 \Rightarrow$ guaranteed diagonalizable! If $\Delta = 0 \Rightarrow$ diagonalizable if and only if **A** is already scalar ($b = c = 0$ and $a = d$).

⚡ Fast-Track 6.2: Rapid 2×2 Eigenvector Construction via Row Adjugate

Once λ is determined, evaluate the first row of $\mathbf{A} - \lambda\mathbf{I}$: $[a - \lambda, \quad b]$. If it is not all zero, an unnormalized eigenvector is immediately obtained by swapping entries and negating one:

$$\mathbf{x} = \begin{vmatrix} -b \\ a - \lambda \end{vmatrix} \quad \text{(since } [a - \lambda, \quad b] \begin{vmatrix} -b \\ a - \lambda \end{vmatrix} = -(a - \lambda)b + b(a - \lambda) = 0\text{)}$$

If the first row is $[0, 0]$, use the second row $[c, \quad d - \lambda] \Rightarrow \mathbf{x} = [-(d - \lambda), \quad c]^T$. Saves 1–2 minutes per eigenvalue by bypassing Gaussian elimination entirely.

⚡ Fast-Track 6.3: Instant Steady-State Markov Distribution without Inverting X

For any 2 × 2 Markov matrix $\mathbf{M} = \begin{vmatrix} 1 - \alpha & \beta \\ \alpha & 1 - \beta \end{vmatrix}$ (where α is probability of leaving state 1 and β is probability of leaving state 2), the steady-state probability vector $\mathbf{u}_\infty = [p, \quad 1 - p]^T$ satisfies the balance equation $\alpha p = \beta(1 - p)$.

The unique normalized steady state is given instantaneously by:

$$\mathbf{u}_\infty = \frac{1}{\alpha + \beta} \begin{vmatrix} \beta \\ \alpha \end{vmatrix}$$

⚡ Fast-Track 6.4: Matrix Powers of Shifted / Rank-1 Perturbations

If a matrix has the structure $\mathbf{A} = \lambda\mathbf{I} + \mathbf{u}\mathbf{v}^T$ (a scalar plus a rank-1 outer product) where $\mathbf{v}^T\mathbf{u} = \gamma$, then $(\mathbf{u}\mathbf{v}^T)^2 = \mathbf{u}(\mathbf{v}^T\mathbf{u})\mathbf{v}^T = \gamma(\mathbf{u}\mathbf{v}^T)$. By the binomial theorem, since $\lambda\mathbf{I}$ and $\mathbf{u}\mathbf{v}^T$ commute:

$$A^k = (\lambda I + uv^T)^k = \lambda^k I + \frac{(\lambda + \gamma)^k - \lambda^k}{\gamma} uv^T$$

Provides immediate closed-form powers for all-ones matrices, correlation matrices, and circulant rank-1 shifts without computing $\mathbf{X}^{-1}$.

⚡ Fast-Track 6.5: Nilpotent Upper-Triangular Toeplitz Exponential

For an $m \times m$ Jordan block $\mathbf{J} = \lambda\mathbf{I} + \mathbf{N}$, because $\lambda\mathbf{I}$ and $\mathbf{N}$ commute, $e^{\mathbf{J}t} = e^{\lambda t\,\mathbf{I}}\, e^{\mathbf{N}t} = e^{\lambda t}\, e^{\mathbf{N}t}$. Because $\mathbf{N}$ is a shift matrix, $e^{\mathbf{N}t} = \sum_{j=0}^{m-1} \frac{(\mathbf{N}t)^j}{j!}$ forms a clean upper-triangular Toeplitz matrix:

$$e^{\mathbf{J}t} = e^{\lambda t} \begin{vmatrix} 1 & t & t^2/2! & \cdots & t^{m-1}/(m-1)! \\ 0 & 1 & t & \cdots & t^{m-2}/(m-2)! \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & t \\ 0 & 0 & \cdots & 0 & 1 \end{vmatrix}$$

⚡ Fast-Track 6.6: Cayley-Hamilton Degree Reduction for High Matrix Powers

By the Cayley-Hamilton Theorem, every matrix satisfies its own characteristic equation: $p(\mathbf{A}) = \mathbf{A}^2 - \text{trace}(\mathbf{A})\mathbf{A} + \det(\mathbf{A})\mathbf{I} = \mathbf{O}$. To compute $\mathbf{A}^k$ without diagonalizing, divide λ^k by $p(\lambda)$:

$$\lambda^k = q(\lambda)\, p(\lambda) + (c_1\,\lambda + c_0) \quad \Rightarrow \quad \mathbf{A}^k = c_1\,\mathbf{A} + c_0\,\mathbf{I}$$

Evaluate at the two eigenvalues λ_1, λ_2 to solve for c_1 and c_0 in seconds:

$$c_1 = \frac{\lambda_1^{\,k} - \lambda_2^{\,k}}{\lambda_1 - \lambda_2}, \qquad c_0 = \frac{\lambda_1\,\lambda_2^{\,k} - \lambda_2\,\lambda_1^{\,k}}{\lambda_1 - \lambda_2}$$

Part 7: Step-by-Step Worked Problems

Problem 6.1: Full Diagonalization & Exact High-Power Evaluation Exam Focus: Spectral Factorization

PROBLEM STATEMENT

Given the matrix $\mathbf{A} = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix}$:

1. Compute the eigenvalues and verify using trace and determinant.
2. Find the corresponding eigenvectors and form the diagonalizing matrix $\mathbf{X}$ and its inverse $\mathbf{X}^{-1}$.
3. Derive a closed-form formula for the general power $\mathbf{A}^k$.
4. Evaluate the exact numerical matrix $\mathbf{A}^{10}$ and verify via trace invariance.

STEP 1: EIGENVALUES VIA FAST-TRACK CHARACTERISTIC POLYNOMIAL

Compute the matrix invariants: trace($\mathbf{A}$) = 1 + 3 = 4, and det($\mathbf{A}$) = (1)(3) - (4)(2) = 3 - 8 = -5.

The characteristic equation is: $p(\lambda) = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \quad \Rightarrow \quad \lambda_1 = 5, \quad \lambda_2 = -1.$

STEP 2: EIGENVECTOR COMPUTATION & MATRIX CONSTRUCTION

For $\lambda_1 = 5$: $\mathbf{A} - 5\mathbf{I}$ = $\begin{vmatrix} -4 & 4 \\ 2 & -2 \end{vmatrix}$. Row 1: $-4x_1 + 4x_2 = 0 \Rightarrow x_1 = x_2$. Choose $\mathbf{x}_1 = \begin{vmatrix} 1 \\ 1 \end{vmatrix}$.

For $\lambda_2 = -1$: $\mathbf{A} - (-1)\mathbf{I} = \mathbf{A} + \mathbf{I} =$ $\begin{vmatrix} 2 & 4 \\ 2 & 4 \end{vmatrix}$. Row 1: $2x_1 + 4x_2 = 0 \Rightarrow x_1 = -2x_2$. Choose $\mathbf{x}_2 = \begin{vmatrix} -2 \\ 1 \end{vmatrix}$.

Assemble $\mathbf{X}$ and compute $\mathbf{X}^{-1}$: $\mathbf{X} = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix}$, $\det(\mathbf{X}) = (1)(1) - (-2)(1) = 3 \neq 0 \Rightarrow \mathbf{X}^{-1} = \frac{1}{3}\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}$.

STEP 3: CLOSED-FORM FORMULA FOR A^K

Using $\mathbf{A}^k = \mathbf{X}\mathbf{\Lambda}^k\mathbf{X}^{-1}$:

$$\mathbf{A}^k = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 5^k & 0 \\ 0 & (-1)^k \end{vmatrix} \frac{1}{3} \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = \frac{1}{3} \begin{vmatrix} 5^k & -2(-1)^k \\ 5^k & (-1)^k \end{vmatrix} \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}$$

Carrying out the final matrix product gives the exact closed form:

$$\mathbf{A}^k = \frac{1}{3} \begin{vmatrix} 5^k + 2(-1)^k & 2 \times 5^k - 2(-1)^k \\ 5^k - (-1)^k & 2 \times 5^k + (-1)^k \end{vmatrix}$$

STEP 4: NUMERICAL EVALUATION AT K = 10 & VERIFICATION

For $k = 10$, note that $5^{10} = 9{,}765{,}625$ and $(-1)^{10} = 1$:

- Top-Left: (9,765,625 + 2) / 3 = 9,765,627 / 3 = **3,255,209**
- Top-Right: (19,531,250 - 2) / 3 = 19,531,248 / 3 = **6,510,416**
- Bottom-Left: (9,765,625 - 1) / 3 = 9,765,624 / 3 = **3,255,208**
- Bottom-Right: (19,531,250 + 1) / 3 = 19,531,251 / 3 = **6,510,417**

$$\mathbf{A}^{10} = \begin{vmatrix} 3{,}255{,}209 & 6{,}510{,}416 \\ 3{,}255{,}208 & 6{,}510{,}417 \end{vmatrix}$$

✔ **Sanity Check:** Trace invariance dictates $\text{trace}(\mathbf{A}^{10}) = \lambda_1^{10} + \lambda_2^{10} = 5^{10} + (-1)^{10} = 9{,}765{,}625 + 1 = \mathbf{9{,}765{,}626}$.
Direct matrix trace: 3,255,209 + 6,510,417 = **9,765,626**. Exact match confirms analytical precision.

Problem 6.2: Coupled Difference Equation: Fibonacci Recurrence & Binet's Formula

Exam Focus: Difference Equations & Companion Matrices

PROBLEM STATEMENT

The Fibonacci sequence is defined by $F_{k+2} = F_{k+1} + F_k$ with initial conditions $F_0 = 0$, $F_1 = 1$.

1. Convert this second-order scalar recurrence into a 2-dimensional first-order matrix system $\mathbf{u}_{k+1} = \mathbf{A}\mathbf{u}_k$.
2. Diagonalize the companion matrix $\mathbf{A}$ and derive Binet's closed formula for F_k.

STEP 1: STATE-SPACE SETUP

Define the state vector $\mathbf{u}_k = \begin{vmatrix} F_{k+1} \\ F_k \end{vmatrix}$. The recurrence relations are:

$$F_{k+2} = F_{k+1} + F_k$$
$$F_{k+1} = F_{k+1} + 0$$

In matrix form:

$$\mathbf{u}_{k+1} = \begin{vmatrix} F_{k+2} \\ F_{k+1} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} F_{k+1} \\ F_k \end{vmatrix} = \mathbf{A}\,\mathbf{u}_k, \quad \text{with} \quad \mathbf{u}_0 = \begin{vmatrix} F_1 \\ F_0 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

STEP 2: EIGENVALUES & EIGENVECTORS OF THE COMPANION MATRIX

Characteristic polynomial: $\det(\mathbf{A} - \lambda\mathbf{I}) = (1 - \lambda)(-\lambda) - (1)(1) = \lambda^2 - \lambda - 1 = 0$.
Using the quadratic formula:

$$\lambda_1 = \frac{1 + \sqrt{5}}{2} = \varphi \approx 1.618034, \quad \lambda_2 = \frac{1 - \sqrt{5}}{2} = -\frac{1}{\varphi} \approx -0.618034$$

Eigenvector equation $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ gives the second row: $x_1 - \lambda x_2 = 0 \Rightarrow x_1 = \lambda x_2$.

Choosing $x_2 = 1$ yields the eigenvectors: $\mathbf{x}_1 = \begin{vmatrix} \lambda_1 \\ 1 \end{vmatrix}$, $\mathbf{x}_2 = \begin{vmatrix} \lambda_2 \\ 1 \end{vmatrix}$.

STEP 3: INITIAL VECTOR DECOMPOSITION & CLOSED-FORM SOLUTION

Expand $\mathbf{u}_0 = c_1\,\mathbf{x}_1 + c_2\,\mathbf{x}_2$:

$$\begin{vmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} c_1 \\ c_2 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

The second row gives $c_1 + c_2 = 0 \Rightarrow c_2 = -c_1$.

Substituting into the first row gives $c_1(\lambda_1 - \lambda_2) = 1$. Since $\lambda_1 - \lambda_2 = \frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = \sqrt{5}$, we have:

$$\mathbf{c_1 = \frac{1}{\sqrt{5}}}, \quad \mathbf{c_2 = -\frac{1}{\sqrt{5}}}$$

Now compute $\mathbf{u}_k = \mathbf{A}^k\,\mathbf{u}_0$:

$$\mathbf{u}_k = \frac{1}{\sqrt{5}}\,\lambda_1^{\,k} \begin{vmatrix} \lambda_1 \\ 1 \end{vmatrix} - \frac{1}{\sqrt{5}}\,\lambda_2^{\,k} \begin{vmatrix} \lambda_2 \\ 1 \end{vmatrix}$$

Extracting the second component (which corresponds to F_k) yields Binet's celebrated formula:

$$\mathbf{F_k = \frac{\lambda_1^{\,k} - \lambda_2^{\,k}}{\sqrt{5}} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right]}$$

✔ **Sanity Check:** Evaluate for $k = 6$: $\varphi^6 \approx 17.944272$, and $\lambda_2^{\,6} = (-0.618034)^6 \approx 0.055728$.
$F_6 = (17.944272 - 0.055728) / \sqrt{5} = 17.888544 / 2.236068 = \mathbf{8.0000}$.
Standard sequence: F_0=0, F_1=1, F_2=1, F_3=2, F_4=3, F_5=5, F_6=8. Exactly verified!

Problem 6.3: Markov System: Urban-Suburban Population Migration

Exam Focus: Markov Matrices & Steady-State Vector

PROBLEM STATEMENT

A demographic study models population migration between a city center and surrounding suburbs each decade. Every 10 years, 20% of urban residents move to the suburbs (80% remain), while 30% of suburban residents move into the city (70% remain).

1. Construct the Markov transition matrix **M** and verify column stochasticity.
2. Find the eigenvalues and eigenvectors of **M**.
3. Given an initial distribution $\mathbf{u}_0 = [0.90,\quad 0.10]^T$ (90% urban, 10% suburban), find the explicit formula for $\mathbf{u}_k$ after k decades and determine the long-term steady-state distribution.

STEP 1: MARKOV MATRIX CONSTRUCTION

Let state vector $\mathbf{u}_k = [u_{urban},\ u_{suburban}]^T$. The transition equations are:

$$u_{urban}(k+1) = 0.80\, u_{urban}(k) + 0.30\, u_{suburban}(k)$$
$$u_{suburban}(k+1) = 0.20\, u_{urban}(k) + 0.70\, u_{suburban}(k)$$

The transition matrix is:

$$\mathbf{M} = \begin{vmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{vmatrix}$$

Column sums: 0.8 + 0.2 = 1.0, and 0.3 + 0.7 = 1.0. All entries non-negative. **M** is a valid Markov matrix.

STEP 2: SPECTRAL ANALYSIS

Invariants: trace(**M**) = 0.8 + 0.7 = 1.5, det(**M**) = (0.8)(0.7) - (0.2)(0.3) = 0.56 - 0.06 = 0.50.

Characteristic polynomial: $\lambda^2 - 1.5\lambda + 0.5 = (\lambda - 1)(\lambda - 0.5) = 0$.
Eigenvalues: $\lambda_1 = 1.0$ (guaranteed steady state), $\lambda_2 = 0.5$ (decaying transient).

Eigenvector for $\lambda_1 = 1$: $(\mathbf{M} - \mathbf{I})\mathbf{x}_1 =$

$$\begin{vmatrix} -0.2 & 0.3 \\ 0.2 & -0.3 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix} \Rightarrow -0.2\, x_1 + 0.3\, x_2 = 0 \Rightarrow \mathbf{x}_1 = \begin{vmatrix} 3 \\ 2 \end{vmatrix}.$$

Normalizing so entries sum to 1 yields the probability steady state: $\mathbf{u}_\infty = \begin{vmatrix} 0.6 \\ 0.4 \end{vmatrix}$.

Eigenvector for $\lambda_2 = 0.5$: $(\mathbf{M} - 0.5\mathbf{I})\mathbf{x}_2 = \begin{vmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix} \Rightarrow x_1 + x_2 = 0 \Rightarrow \mathbf{x}_2 = \begin{vmatrix} 1 \\ -1 \end{vmatrix}.$

STEP 3: INITIAL VECTOR EXPANSION & LONG-TERM LIMIT

Expand $\mathbf{u}_0 = [0.9,\ \ 0.1]^T$ in terms of the eigenvectors:

$$\begin{vmatrix} 0.9 \\ 0.1 \end{vmatrix} = c_1 \begin{vmatrix} 0.6 \\ 0.4 \end{vmatrix} + c_2 \begin{vmatrix} 1 \\ -1 \end{vmatrix}$$

Summing the components: $(0.6 + 0.4)c_1 + (1 - 1)c_2 = 0.9 + 0.1 = 1.0 \Rightarrow c_1 = 1.0$.

Then $0.6(1.0) + c_2 = 0.9 \Rightarrow c_2 = 0.30$.

The exact state after k decades is:

$$\mathbf{u}_k = \mathbf{M}^k \mathbf{u}_0 = (1.0)(1)^k \begin{vmatrix} 0.6 \\ 0.4 \end{vmatrix} + (0.3)(0.5)^k \begin{vmatrix} 1 \\ -1 \end{vmatrix} = \begin{vmatrix} 0.6 + 0.3(0.5)^k \\ 0.4 - 0.3(0.5)^k \end{vmatrix}$$

As $k \to \infty$, $(0.5)^k \to 0$, leaving:

$$\mathbf{u}_\infty = \begin{vmatrix} \mathbf{0.60} \\ \mathbf{0.40} \end{vmatrix} \quad \textbf{(60\% Urban, 40\% Suburban)}$$

✓ **Sanity Check:** Evaluate after 1 decade ($k = 1$): $\mathbf{u}_1 = [0.6 + 0.15,\ \ 0.4 - 0.15]^T = [0.75,\ \ 0.25]^T$.

Direct matrix product: $\mathbf{M}\,\mathbf{u}_0 = \begin{vmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{vmatrix} \begin{vmatrix} 0.9 \\ 0.1 \end{vmatrix} = \begin{vmatrix} 0.72 + 0.03 \\ 0.18 + 0.07 \end{vmatrix} = \begin{vmatrix} 0.75 \\ 0.25 \end{vmatrix}$. Perfect agreement!

Problem 6.4: System of Linear Differential Equations via Matrix Exponential Exam Focus: Continuous Systems & Matrix Exponential

PROBLEM STATEMENT

Solve the initial value problem $du/dt = \mathbf{A}u$ with $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ and $\mathbf{u}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

1. Determine the eigenvalues and evaluate asymptotic stability.
2. Compute the matrix exponential $e^{\mathbf{A}t}$.
3. Find the explicit solution $\mathbf{u}(t)$ and verify at $t = 0$.

STEP 1: EIGENVALUES & STABILITY CLASSIFICATION

Invariants: trace($\mathbf{A}$) = 0 + (-3) = -3, det($\mathbf{A}$) = (0)(-3) - (1)(-2) = 2.

Characteristic equation: $\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0 \Rightarrow \lambda_1 = -1, \quad \lambda_2 = -2$.

Since both eigenvalues are strictly negative real numbers ($\lambda_2 < \lambda_1 < 0$), the origin is an **asymptotically stable nodal sink**: all trajectories decay exponentially to $\mathbf{0}$ as $t \to \infty$.

STEP 2: EIGENVECTORS & DIAGONALIZATION FACTORIZATION

For $\lambda_1 = -1$: $\mathbf{A} - (-1)\mathbf{I} = \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \Rightarrow x_1 + x_2 = 0 \Rightarrow \mathbf{x}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

For $\lambda_2 = -2$: $\mathbf{A} - (-2)\mathbf{I} = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} \Rightarrow 2x_1 + x_2 = 0 \Rightarrow \mathbf{x}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Form $\mathbf{X}$ and invert: $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$, $\det(\mathbf{X}) = -2 - (-1) = -1 \Rightarrow \mathbf{X}^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$.

STEP 3: CONSTRUCT $e^{\mathbf{A}t}$ AND FIND EXPLICIT TRAJECTORY

Using $e^{\mathbf{A}t} = \mathbf{X}\, e^{\mathbf{\Lambda}t}\, \mathbf{X}^{-1}$:

$$e^{\mathbf{A}t} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

Multiplying by $\mathbf{u}(0) = [1, \quad 0]^T$ extracts the first column of $e^{\mathbf{A}t}$:

$$\mathbf{u}(t) = e^{\mathbf{A}t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} \end{bmatrix}$$

✓ **Sanity Check:** Check $t = 0$: $\mathbf{u}(0) = [2 - 1, \quad -2 + 2]^T = [1, \quad 0]^T$. Correct.

Check velocity vector at $t = 0$: $du/dt|_{t=0} = [-2e^{-t} + 2e^{-2t}, \quad 2e^{-t} - 4e^{-2t}]^T|_{t=0} = [0, \quad -2]^T$.

Matrix multiplication: $\mathbf{A}\,\mathbf{u}(0) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$. Exactly matches ODE $du/dt = \mathbf{A}u$!

Problem 6.5: Defective Matrix Analysis & Jordan Form Construction Exam Focus: Defective Matrices & Jordan Canonical Form

PROBLEM STATEMENT

Consider the matrix $\mathbf{A} = \begin{vmatrix} 4 & 1 \\ -1 & 2 \end{vmatrix}$.

1. Find the eigenvalues and determine their algebraic and geometric multiplicities. Prove that **A** is defective.
2. Construct a generalized eigenvector and form the similarity matrix **M** such that $\mathbf{M}^{-1}\,\mathbf{A}\,\mathbf{M} = \mathbf{J}$ is in Jordan Canonical Form.
3. Compute the matrix exponential $e^{\mathbf{A}t}$.

STEP 1: PROOF OF DEFECTIVENESS

Invariants: trace(**A**) = 4 + 2 = 6, det(**A**) = (4)(2) - (1)(-1) = 8 + 1 = 9.

Characteristic polynomial: $p(\lambda) = \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0$.
There is a single repeated eigenvalue $\lambda = 3$ with **AM = 2**.

Eigenspace null(**A** - 3**I**): **A** - 3**I** = $\begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix}$. The rank is 1. Thus:

$$\mathbf{GM(\lambda = 3) = 2 - rank(A - 3I) = 2 - 1 = 1}$$

Since **GM(3) = 1 < AM(3) = 2**, the matrix **A** is missing one eigenvector: it is **defective** and non-diagonalizable.

STEP 2: CONSTRUCTING THE GENERALIZED EIGENVECTOR CHAIN & MATRIX M

Find the true eigenvector $\mathbf{v}_1$ by solving $(\mathbf{A} - 3\mathbf{I})\mathbf{v}_1 = \mathbf{0} \Rightarrow v_{1,1} + v_{1,2} = 0$. Choose $\mathbf{v}_1 = \begin{vmatrix} 1 \\ -1 \end{vmatrix}$.

Next, find a generalized eigenvector $\mathbf{v}_2$ by solving the chain equation $(\mathbf{A} - 3\mathbf{I})\mathbf{v}_2 = \mathbf{v}_1$:

$$\begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} \begin{vmatrix} v_{2,1} \\ v_{2,2} \end{vmatrix} = \begin{vmatrix} 1 \\ -1 \end{vmatrix} \quad \Rightarrow \quad v_{2,1} + v_{2,2} = 1$$

Pick the simplest particular solution: $\mathbf{v}_2 = \begin{vmatrix} 1 \\ 0 \end{vmatrix}$.

Assemble the Jordan basis matrix $\mathbf{M} = [\mathbf{v}_1 \quad \mathbf{v}_2]$ and compute its inverse: $\mathbf{M} = \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix}$, $\det(\mathbf{M}) = 0 - (-1) = 1 \Rightarrow \mathbf{M}^{-1} = \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix}$.

Verify similarity to Jordan form:

$$\mathbf{M}^{-1}\,\mathbf{A}\,\mathbf{M} = \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 4 & 1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 3 & 4 \\ -3 & -1 \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ 0 & 3 \end{vmatrix} = \mathbf{J}$$

STEP 3: EVALUATING THE DEFECTIVE MATRIX EXPONENTIAL

For the Jordan block $\mathbf{J} = 3\mathbf{I} + \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}$, the exponential is $e^{\mathbf{J}t} = e^{3t} \begin{vmatrix} 1 & t \\ 0 & 1 \end{vmatrix}$. Using $e^{\mathbf{A}t} = \mathbf{M}\, e^{\mathbf{J}t}\, \mathbf{M}^{-1}$:

$$e^{\mathbf{A}t} = e^{3t} \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} \begin{vmatrix} 1 & t \\ 0 & 1 \end{vmatrix} \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = e^{3t} \begin{vmatrix} 1+t & t \\ -t & 1-t \end{vmatrix}$$

✔ **Sanity Check:** Evaluate the derivative at $t = 0$: $\frac{d}{dt}(e^{\mathbf{A}t})\big|_{t=0}$.

Using product rule: $3\, e^{3t} \begin{vmatrix} 1+t & t \\ -t & 1-t \end{vmatrix} + e^{3t} \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix}$ at $t = 0$ yields $3 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} = \begin{vmatrix} 4 & 1 \\ -1 & 2 \end{vmatrix} = \mathbf{A}$. Verified!

Part 8: Comprehensive Spectral Taxonomy & Quick-Reference Matrix

Matrix Class	Defining Property	Eigenvalues λ_i	Eigenvectors & Diagonalizability	Dynamical Power Action A^k
Diagonalizable	GM(λ) = AM(λ) for all λ	Arbitrary in $\mathbb{C}$	*n* linearly independent eigenvectors ($\mathbf{A} = \mathbf{X}\Lambda\mathbf{X}^{-1}$)	$\mathbf{A}^k = \mathbf{X}\,\mathrm{diag}(\lambda_i^k)\,\mathbf{X}^{-1}$
Symmetric (Vol 07)	$\mathbf{S}^T = \mathbf{S}$	Strictly real ($\lambda_i \in \mathbb{R}$)	Orthonormal eigenvectors ($\mathbf{S} = \mathbf{Q}\Lambda\mathbf{Q}^T$, never defective)	Rank-1 projections $\sum \lambda_i^k \mathbf{q}_i\mathbf{q}_i^T$
Orthogonal	$\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$	Unit circle $\lvert\lambda_i\rvert = 1$	Unitary orthonormal eigenvectors	Rigid isometric rotation / reflection
Markov	$m_{ij} \geq 0$, $\sum_i m_{ij} = 1$	$\lambda_1 = 1$, all others $\lvert\lambda_i\rvert \leq 1$	Steady-state vector $\mathbf{u}_\infty = \mathbf{x}_1$ ($\lambda_1 = 1$)	Converges to stationary distribution
Nilpotent	$\mathbf{N}^p = \mathbf{O}$	All $\lambda_i = 0$	Defective (only GM ≥ 1 true eigenvector)	Extinct: $\mathbf{N}^k = \mathbf{O}$ for all $k \geq p$
Defective	GM(λ) < AM(λ) for some λ	Repeated eigenvalues	Lacks full eigenbasis ⇒ Jordan Form $\mathbf{A} = \mathbf{M}\mathbf{J}\mathbf{M}^{-1}$	Coupled polynomial-exponential modes $t^p\lambda^k$

◆ The Core Pedagogical Bridge to Volume 07 & Volume 08

In this Volume 06, we established that a general square matrix is diagonalizable if and only if its geometric multiplicities match its algebraic multiplicities (GM = AM). When GM < AM, the matrix is defective and requires the Jordan form with nilpotent shear blocks.

In **Volume 07 (Symmetric Matrices & The Spectral Theorem)**, we uncover the most remarkable theorem in linear algebra: whenever a matrix is real and symmetric ($\mathbf{S} = \mathbf{S}^T$), defectiveness is completely impossible. Every symmetric matrix is guaranteed to possess real eigenvalues and a complete set of mutually orthogonal eigenvectors, leading to the orthogonal decomposition $\mathbf{S} = \mathbf{Q}\Lambda\mathbf{Q}^T$.

In **Volume 08 (Positive Definite Matrices & Quadratic Forms)**, we restrict symmetric matrices to the positive regime ($\lambda_i > 0$), unlocking convex energy surfaces, Cholesky factorization, hyper-ellipsoids, and optimization principles.